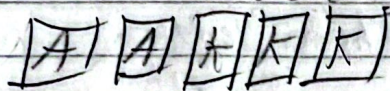


$${}_nC_r = \frac{n!}{r!(n-r)!} \quad {}_nP_r = \frac{n!}{(n-r)!}$$

Combinatorial Probability

- P = function that is a set of all possible events on S
- ① $\forall A \quad P(A) \geq 0$ ② $P(S) = 1$ ③ $A_1, A_2, \dots, A_i \quad A_i \cap A_j = \emptyset$
- Classical definition: $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ for finite sample space
- Sample space is finite
- All outcomes are equally likely $P(S_i) = P(S_j) \quad \forall i, j$

Ex1 Deck of cards: 5 cards are dealt Find P that 2 are Aces and 3 are Kings



order doesn't matter

$$P(4 \text{ Aces choose } 2) \cdot P(4 \text{ Kings choose } 3) = \frac{\binom{4}{2} \cdot \binom{4}{3}}{\binom{52}{5}} \approx 0.0009\%$$

End

2 Rules: Multiplication Rule, Summation Rule

permutations w/ repetition: n_1 repeats, n_2 repeats, ..., n_k repeats

$$\text{number of permutations } N = \frac{n!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

Ex2 Find P that exactly 3 cards are hearts out of 5 dealt

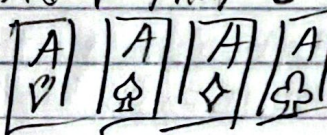
$$13 \text{ H and } 39 \text{ not H } (\binom{13}{3} \cdot \binom{39}{2}) / \binom{52}{5}$$

Find p that at least 3 are Hearts: 3 or 4 or 5

$$\frac{\binom{13}{3} \cdot \binom{39}{2} + \binom{13}{4} \cdot \binom{39}{1} + \binom{13}{5} \cdot \binom{39}{0}}{\binom{52}{5}}$$

End

Ex Find p that 2 are aces and 3 are hearts (not mutually exclusive)



Ace not heart $\binom{3}{2} \cdot \binom{12}{3}$

Ace w/ heart $1 \cdot \binom{3}{1} \cdot \binom{12}{2} \cdot \binom{36}{1}$ extra card

$$\frac{\binom{3}{2} \binom{12}{3} + 1 \cdot \binom{3}{1} \binom{12}{2} \binom{36}{1}}{\binom{52}{5}}$$

End

Ex Birthday Problem: 34 students, chance of 2 sharing a birthday

$1 - P(\text{no one})$ 365 students 34 students $365 P_{34} = \text{All born different days}$

$$1 - \frac{365 P_{34}}{365^{34}} \approx 80\%$$

$$8 = 365^{34}$$